

PLANCKS 2023 Problems Booklet

Dear contestants Welcome to PLANCKS 2023

- The language used in the competition is English.
- The contest consists of **10 exercises**, **each worth 10 points**. Subdivisions of points are indicated in the exercises.
- All exercises have to be handed in on a separate sheet of paper. Please mark each sheet with your team name and the exercise number. **Marking each sheet is necessary for an exercise to be scored**.
- When a problem is unclear, a participant can ask, via the crew, for clarification. If the response is relevant to all teams, the jury will provide this information to the other teams.
- You are allowed to use a dictionary: English to your native language.
- You are allowed to use a non-programmable, not-graph calculator (But scientific is okay).
- No books or other sources, except for this exercise booklet and a dictionary, are to be consulted during the competition.
- The jury has the right to disqualify teams for misbehavior or breaking the rules.
- The use of hardware (including phones, tablets etc.) is not approved, except for watches and medical equipment. Please leave your phones in an envelope.
- In situations to which no rule applies, the OC decides.

May the best physics team win!

Milan, 14/05/2023

1 Quantum Mechanics

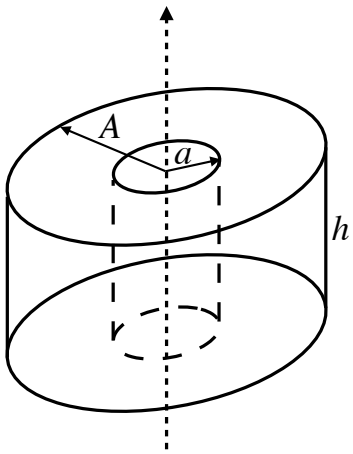


Figure 1: A cavity between two coaxial cylinders.

Find the energetic levels of a particle living in a cavity between two coaxial cylinders of height h and radii a, A ($a < A$), as depicted in Fig. 1. Consider the following cases:

i) **(1 point)** $h \gg A$ (both cylinders extend indefinitely along their axis);

ii) **(3 points)** The cavity is closed by two lids on the top and bottom ends;

iii) **(6 points)** A uniform magnetic field $\vec{B}$, directed along the cylinder axis, is present only inside the smallest cylinder, but is zero anywhere else (including in the cavity).

For the radial part, it is enough to obtain the equation(s) whose solutions give the energy levels.

Hints:

i) The gradient, curl and Laplacian in cylindrical coordinates are

$$\vec{\nabla} = \frac{\partial}{\partial z}\hat{z} + \frac{\partial}{\partial r}\hat{r} + \frac{1}{r}\frac{\partial}{\partial \phi}\hat{\phi}\,, \tag{1}$$

$$\vec{\nabla}\times\vec{A} = \frac{1}{r}\left(\frac{\partial(rA_\phi)}{\partial r} - \frac{\partial A_r}{\partial \phi}\right)\hat{z} + \left(\frac{1}{r}\frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z}\right)\hat{r} + \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r}\right)\hat{\phi}\,, \tag{2}$$

$$\nabla^2 = \frac{\partial^2}{\partial z^2} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2}{\partial \phi^2}\,. \tag{3}$$

ii) The Bessel functions are the solutions of the differential equation

$$x^2y'' + xy' + (x^2 - \alpha^2)y = 0 \quad \alpha > 0\,. \tag{4}$$

The Bessel functions of the first kind, $y = J_\alpha(x)$, are regular at $x \to 0$, while those of the second kind, $y = Y_\alpha(x)$, diverge in the same limit. In Fig. 2 some of the Bessel functions are plotted.

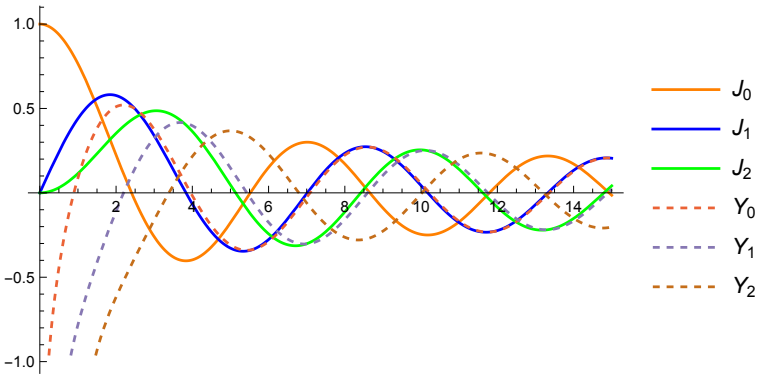


Figure 2: Bessel functions of the first and second kind.

2 Thermodynamics

Let's consider a spherical soap bubble containing n moles of a perfect gas, whose conductive surface has a charge q. The bubble is at atmospheric conditions: let T_0 be the temperature and p_0 the pressure, while γ_0 is the constant of surface tension of the **uncharged** water and soap layer.

i) **(2 points)** Show that the constant of surface tension depends on the charge;

ii) **(4 points)** Determine the equilibrium conditions for the charged bubble. Verify that there is always one and only one value that satisfies these conditions;

iii) **(4 points)** Study the stability of the equilibrium.

3 Quantum Optics

Rabi oscillations — A traveling two-level atom interacts with a classical single-mode oscillating field inside a lossless microwave cavity. The detuning between the atomic transition and the cavity field is $\Delta\omega$ and the Hamiltonian describing the atom-cavity field system in the interaction picture is given by:

$$H_{\rm int} = \hbar\frac{\Delta\omega}{2}\sigma_3 - \hbar\frac{\Omega_0}{2}\sigma_1\,, \tag{41}$$

where $\sigma_3 = |e\rangle\langle e| - |g\rangle\langle g|$ and $\sigma_1 = |e\rangle\langle g| + |g\rangle\langle e|$ are Pauli operators, $|e\rangle$ and $|g\rangle$ refer to the excited and ground state of the atom, respectively, and Ω_0 is the Rabi frequency.

Assuming that the atom is initially in the ground state, namely, $|\psi_0\rangle = |g\rangle$:

i) **(2 points)** find the evolved state:

$$|\psi_t\rangle = \exp\left(-i\frac{H_{\rm int}\,t}{\hbar}\right)|\psi_0\rangle. \tag{42}$$

Then, write the analytic expression of the probability of finding the atom in the excited state $P_e(t) = |\langle e|\psi_t\rangle|^2$ and comment the result in the following cases:

ii) **(1 point)** when $\Delta\omega = 0$ (resonance);

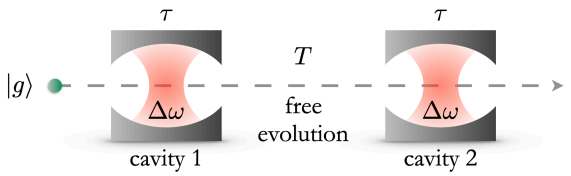
iii) **(1 point)** when $\Delta\omega \gg \Omega_0$ (large detuning).

If the interaction time is set to $t = \pi/\Omega_0$ (corresponding to a π-pulse) and $\Delta\omega \neq 0$:

iv) **(1 point)** write $P_e(t = \pi/\Omega_0)$ as a function of the quantity $x = \Delta\omega/\Omega_0$ and find the value $x > 0$ at which the first minimum occurs.

Ramsey fringes — Now we consider a setup involving two cavities, as displayed in the figure.

The interaction between the atom and the field inside each cavity is still described by the Hamiltonian (41), whereas, during the free evolution between the two cavities, the Hamiltonian reads (in the interaction picture):



$$H_{\rm free} = \hbar\frac{\Delta\omega}{2}\sigma_3. \tag{43}$$

We also assume that the two cavity fields have the same phase that we set equal to zero. The interaction time inside both the cavities is set to $\tau = \pi/(2\,\Omega_0)$, corresponding to a $\frac{\pi}{2}$–pulse, and the free evolution lasts for a time $T = \xi\tau$, with $\xi \in \mathbb{N}$.

If the initial state of the atom is still $|\psi_0\rangle = |g\rangle$ and the evolved state after the whole evolution (cavity 1 — free evolution — cavity 2) is $|\psi_{\rm out}\rangle$:

v) **(2 points)** prove that the probability of finding the atom in the excited state $\tilde{P}_e = |\langle e|\psi_{\rm out}\rangle|^2$ as a function of $x = \Delta\omega/\Omega_0$ can be written as:

$$\tilde{P}_e(x;\xi) = g(x)\, f_\xi(x), \tag{44}$$

where:

$$g(x) = \frac{4}{1+x^2}\,\sin^2\left(\frac{\pi}{4}\sqrt{1+x^2}\right),$$

and

$$f_\xi(x) = \left[\cos\left(\frac{\pi}{4}\xi x\right)\cos\left(\frac{\pi}{4}\sqrt{1+x^2}\right) - \frac{x}{\sqrt{1+x^2}}\sin\left(\frac{\pi}{4}\xi x\right)\sin\left(\frac{\pi}{4}\sqrt{1+x^2}\right)\right]^2,$$

vi) **(3 points)** show that the smallest value $x_{\rm m} > 0$ such that $\tilde{P}_e(x_{\rm m};\xi) = 0$ scales as $1/\xi$ in the limit $\xi \gg 1$.

4 Structure of matter

Counting Spin-polarized Electrons - Consider an electron gas at temperature $T = 0$. Initially neglect the Coulombic electron-electron interaction. Assume that this electron gas is immersed in a static uniform magnetic field $B = 8.64$ T coupled to the electron spins (i.e. neglect the field effect on the orbital motion of the electrons). Let the length quantity L have the following value: $L = 41.1$ nm.

i) **(0.5 points)** Derive an expression for the maximum number density $n = N/V$ such that in the ground state the electron-spin magnetic moments of all N electrons align to the magnetic field, and evaluate the value of this density (in m^{-3}).

ii) **(1 point)** Describe the electrons of a metallic nanoparticle adopting a radically simplified model: a cubic volume $V = L \times L \times L$. Evaluate $N_{\max} = n\,V$ using the result of the previous question.

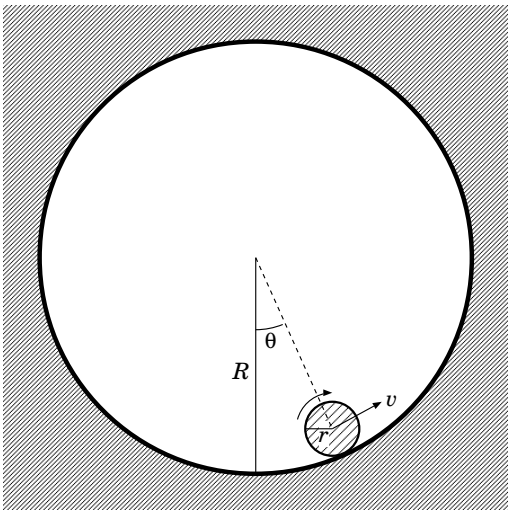
iii) **(1 point)** Apply periodic boundary conditions (PBC) to the free-electron wave functions in the $L \times L \times L$ box. By taking the resulting discrete single-electron energy levels and degeneracies into account, what is the actual maximum number $N_{\max}^{\mathrm{PBC}}$ that are 100% spin-aligned in the same magnetic field?

iv) **(2 points)** Modify the nanoparticle model to a cubic-shaped $L \times L \times L$ infinitely-deep potential well, thus replacing the PBC of question iii) with null boundary conditions (NBC) to the electron wave functions. By taking the resulting discrete single-electron energy levels and degeneracies into account, what maximum number $N_{\max}^{\mathrm{NBC}}$ of electrons with 100% spin alignment (in the same magnetic field) does this model yield?

v) **(2.5 points)** Assume that (at a Hartree-Fock mean-field level) the electric repulsion between the electrons is compensated by their attraction to a uniform background of positive charge. In this effective independent-electrons picture, for the residual exchange effect of the electron-electron Coulomb interaction, take the following simplified Hamiltonian: $H_{\mathrm{exch}} = -\frac{1}{2}J(N_\uparrow^2 + N_\downarrow^2)^{1/2}$. Verify that, assuming $J > 0$, this term favors spin-aligned states relative to spin-antiparallel states. Estimate the value of J by the Coulomb repulsion of two uniformly delocalized electrons in the $L \times L \times L$ box.

vi) **(3 points)** In the continuum model of question ii), how does the presence of the exchange term H_{exch} modify the maximum number $N_{\max}$ of electrons that are 100% spin-aligned in the same magnetic field? To what value $N_{\max}^{\mathrm{exch}}$?

5 Classical Mechanics



A ball, with spherical symmetry, with external radius r, mass m, has a moment of inertia with respect to its center-of-mass (CM) equal to $I^* = \beta\, mr^2$. The ball rolls without slipping ("pure rolling") inside a spherical cavity of radius R. Everything is embedded in the gravitational field g at Earth's surface. Rolling frictions are negligible. The contact point of the ball on the spherical surface is displaced with respect to the lowest point by an angle θ, measured from the center of the spherical cavity.

Initially, the ball is located in the lowest point of the cavity and moves with a CM speed v_0.

In general, a pure rolling motion on a spherical surface has 2 degrees of freedom (DoF), that could be parameterized by a polar angle θ and an azimuthal angle ϕ.

i) **(2 points)** Write down the Lagrangian of the rolling ball as a function of θ, $\dot{\theta}$, ϕ, $\dot{\phi}$ and prove that in this specific problem only θ matters, i.e. that the problem has only 1 DoF.

(In case you fail proving the previous point, just assume it's true and continue...)

Write down, as functions of θ and of the other parameters of the problem, the following formulae:

ii) **(0.5 points)** the speed of the CM: $v(\theta) = \ldots$

iii) **(0.5 points)** the tangent acceleration of the CM: $a_t(\theta) = \ldots$

iv) **(0.5 points)** the reaction of the constraint, produced by the surface: $F_{constr}(\theta) = \ldots$

v) **(0.5 points)** the static friction, produced by the surface: $F_{frict}(\theta) = \ldots$

Keeping in mind the spherical symmetry of the rolling object:

vi) **(2 points)** what is the maximum possible value of β ($\beta_{\max}$)?

Now, let's assume $\beta = \beta_{\max}$, $r = 1$ cm, $R = 0.5$ m, $\mu_s = 0.9$:

vii) **(4 points)** what is the minimum value $v_0^{\min}$ of the initial speed v_0, such that the rolling ball keeps the conditions of "pure rolling" during all its motion? [provide both formula and numerical value]

6 Electromagnetism

A coaxial cable (Figure 4a) consists of a central *ohmic* conductor of radius a, length h, resistance R surrounded by a coaxial perfectly conducting cylinder of radius b; the region between the central conductor and the external cylinder is free space.

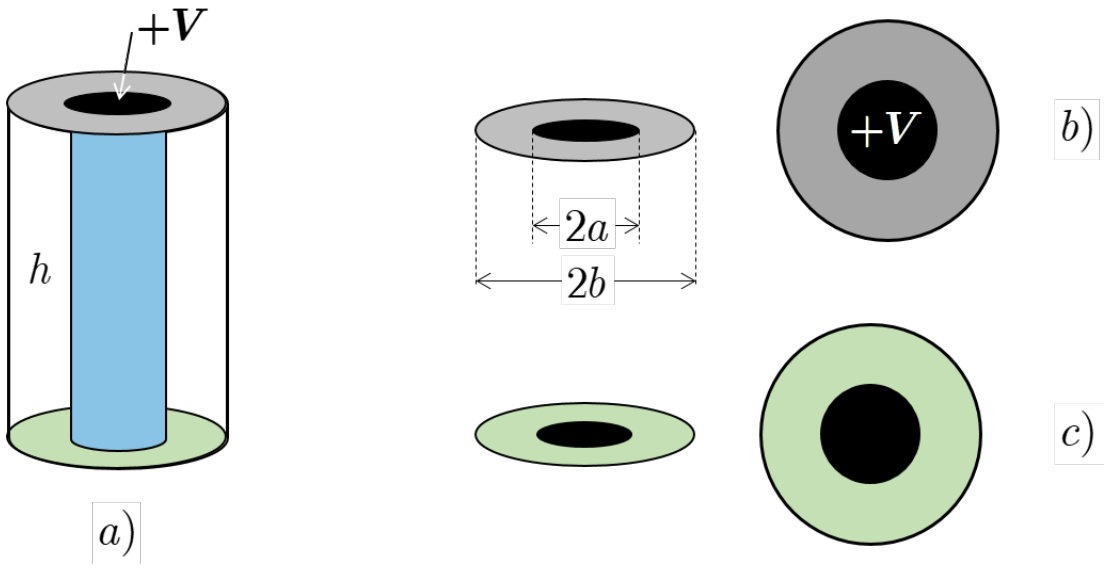


Figure 4: geometry of the battery a), coaxial cable b), and resistor c)

The cable is connected from on one side to a circular battery of negligible thickness (Figure 4b); the region $r < a$ is at potential V, the potential inside the battery decreasing radially to zero as

$$\phi(r) = \frac{V}{\ln \frac{a}{b}} \ln \frac{r}{b}$$

On the other side, the cable is connected to a circular load resistor R_L, constructed with *ohmic* conductor, of negligible thickness, of inner and outer radii a and b respectively (Figure 4c). Consider the four regions

$$\text{region 1} \quad \begin{matrix} 0 < r < a \\ 0 < z < h \end{matrix} \qquad \text{region 2} \quad \begin{matrix} a < r < b \\ z = 0 \end{matrix}$$

$$\text{region 3} \quad \begin{matrix} a < r < b \\ z = h \end{matrix} \qquad \text{region 4} \quad \begin{matrix} a < r < b \\ 0 < z < h \end{matrix}$$

i) **(4 points)** Determine the electric field in each of the four regions and the surface charge densities on the boundaries separating the four regions. *Hint:* solve the Laplace equation in cylindrical coordinates (see also the first hint of problem 1) using variables separation method with separation constant equal to 0.

ii) **(2 points)** Determine the Poynting vector in the above regions and discuss the energy fluxes among the various regions.

iii) **(4 points)** Finally, assume $R = 0$ and determine the electromagnetic momentum of the system and compare it with movements of the parts of the system, if any.

7 Medical Physics

In conventional external beam radiation therapy, X-rays are used to impart a radiation dose to the tumour, sparing the surrounding healthy tissue as much as possible. Consider a medical linear accelerator (linac) where electrons are accelerated to an energy of 15 MeV and then stopped in a metal target. A portion of their kinetic energy is transformed into bremsstrahlung X-rays that form a radiotherapy photon beam. Because of the energies of the produced X-rays and the low atomic number of tissues, the most important interaction of the photon beam with patient tissues is Compton scattering.
Consider a Compton effect interaction where a photon is scattered on a free (i.e. loosely bound) electron with a scattering angle θ (i.e. angle between the incident photon direction and the scattered photon direction).

i) **(1 point)** Derive the expression of the kinetic energy of the recoil electron as a function of the incident photon energy and of the scattering angle θ;

ii) **(1 point)** Show that the energy of photon scattered with angles θ larger than $\pi/2$ cannot exceed 511 keV no matter how high the incident photon energy is.

Consider now the interaction of the X-rays produced by the linac with an absorber, and the related interaction coefficients μ/ρ (mass attenuation coefficient) and μ_{en}/ρ (mass energy absorption coefficient).

iii) **(1 point)** Express the ratio $\frac{\mu_{en}/\rho}{\mu/\rho}$ in terms of: (a) the mean fraction f_j of the incident photon energy that is transferred to kinetic energy of charged particles in an interaction of type j, (b) the component cross section σ_j relating to the interaction of type j, and (c) the fraction g of the kinetic energy transferred by photons to charged particles that is subsequently lost in radiative processes.

The ratio $\frac{\mu_{en}/\rho}{\mu/\rho}$ vs. photon beam energy produced by the linac, for a high Z absorber (lead), is plotted in Figure 6.

iv) **(1 point)** How do you explain the minimum value of the ratio $\frac{\mu_{en}/\rho}{\mu/\rho}$ in lead around 100 keV?

v) **(1 point)** How do you explain the decrease of the ratio $\frac{\mu_{en}/\rho}{\mu/\rho}$ in lead with increasing the photon energy occurring above approximately 10 MeV?

Accelerated charged particles, with mass $m \gg m_{electron}$, can also be used for external beam radiotherapy to treat tumours. The advantage of hadrontherapy localised dose deposition with respect to conventional radiotherapy is related to the typical energy release of such particles in matter, which follows the Bethe-Bloch equation. The presence of the Bragg peak, that appears when the heavy charged particle velocity approaches to zero, defines also the concept of *range* of such particles in matter.

vi) **(1 point)** Describe the main dependencies of hadrons energy loss in matter, discussing the characteristics of the Bragg peak (why it is a peak and why it has a non-null width).

vii) **(1 point)** Express the range scaling laws in the case of:

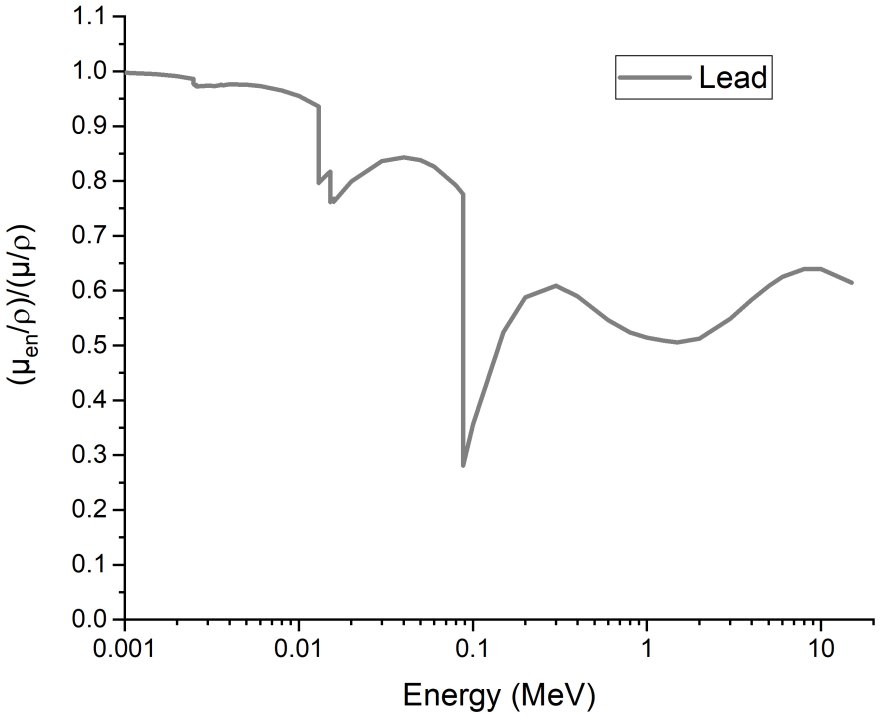


Figure 6

a) different particles with same velocity, traversing the same medium;

b) same particle traversing different media.

viii) **(1 point)** Compute the ratio of protons range with respect to carbon ions range, when protons and carbon ions have the same velocity.

Charged particles undergo also multiple coulomb scattering (MCS) by the nuclei field when traversing a medium, causing small deflections from the primary track path.

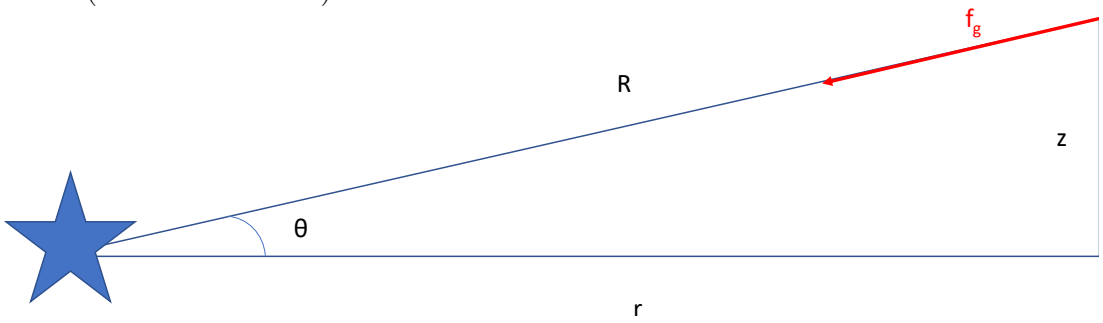
ix) **(1 point)** Describe the expression of the root mean squared deflection angle projected to a plane (θ_0) due to the MCS in the case of small traversed thickness.

x) **(1 point)** Considering a 150 MeV proton and a 285 MeV/u carbon ion, compute the ratio of θ_0 of proton over carbon ion after 1 meter of air.

8 Astrophysics of protoplanetary discs

Stars form from gas gravitational collapse in molecular clouds. As a result of angular momentum conservation, the collapse also produces discs, namely dense structures concentrated around a specific plane (often called the disc mid-plane), which rotate around the newly formed star. It is in these discs that planets form and for this reason they are often called proto-planetary discs. Given the geometry of these systems, it is convenient to use a cylindrical coordinate system to describe them.

IMPORTANT: pay attention to the difference between cylindrical radius r and spherical radius R (see sketch below).



i) **(3 points)** Assume for simplicity that the disc is vertically isothermal. In this case the gas pressure P is simply given by $P = c_s^2 \rho$, where c_s is the gas sound speed and ρ is the gas density. After further assuming that the gas is in hydrostatic equilibrium and so there is no motion along the vertical direction, derive the vertical density profile $\rho(z)$. To do so you can neglect the disc self-gravity and assume that only the star contributes to the gravitational force.

ii) **(1 point)** These discs are typically geometrically thin, meaning that their thickness H is much smaller than the radial coordinate r: $H \ll r$. Simplify the expression you derived in the previous question under this hypothesis and derive a relation between the gas density in the midplane $\rho_0(r) = \rho(r, z = 0)$ and the disc surface density, defined as $\Sigma = \int_{-\infty}^{+\infty} \rho(z) \mathrm{d}z$, that is, the vertical integral of the disc density.

iii) **(3 points)** Considering only the disc midplane at $z = 0$, derive the rotation curve $v_\phi(r)$ of the disc. As before, you can neglect the disc self-gravity and assume that only the star contributes to the gravitational field. In addition to gravity, you need to take into account gas pressure. For simplicity, you can assume that the gas density in the midplane is a power-law function of the radial coordinate: $\rho_0 \propto r^{-p}$. In the same way, you can also assume that the sound speed $c_s(r) \propto r^{-q}$ is a power-law.

iv) **(1 point)** Proto-planetary discs are close to Keplerian rotation. Using the expression you derived in the previous point, quantify the fractional deviation of the rotation curve from Keplerian rotation, and show that this is indeed small provided that the disc is thin.

v) **(2 points)** Generalise the expression you derived for the rotation curve for $z \neq 0$. As in question 2, use the thin disc approximation and only consider the case of small z/r. Does gas at high altitude rotate faster or slower than the gas in the midplane?

Hints:
The operator $(\mathbf{A}\cdot\nabla)\mathbf{B}$ in cylindrical coordinates is the following:

$$
\begin{aligned}
(\mathbf{A}\cdot\nabla)\mathbf{B} = &\left(A_r\frac{\partial B_r}{\partial r} + \frac{A_\phi}{r}\frac{\partial B_r}{\partial \phi} + A_z\frac{\partial B_r}{\partial z} - \frac{A_\phi B_\phi}{r}\right)\hat{\mathbf{r}} + \\
&+\left(A_r\frac{\partial B_\phi}{\partial r} + \frac{A_\phi}{r}\frac{\partial B_\phi}{\partial \phi} + A_z\frac{\partial B_\phi}{\partial z} + \frac{A_\phi B_r}{r}\right)\hat{\phi} + \\
&+\left(A_r\frac{\partial B_z}{\partial r} + \frac{A_\phi}{r}\frac{\partial B_z}{\partial \phi} + A_z\frac{\partial B_z}{\partial z}\right)\hat{\mathbf{z}}
\end{aligned}
$$

9 Particle physics

Magnetars, namely a type of neutron stars with an extremely high magnetic field, are good candidates to be the sources of ultra-high energy cosmic rays, which are the most energetic particles ever observed. However, it's likely that they can accelerate up to the highest energies ($\sim$ 100 EeV) only in the very early stages of their life. One of the closest known Magnetars is SWIFT J1818.0-1607, which is estimated to be only 240 years old. It is located 4.4 kpc away from the Earth, on the galactic plane. Suppose SWIFT J1818.0-1607 emitted for a very small time at its birth cosmic rays, and that they consist only in three components: protons (P), neutrons (N) and iron nuclei (I), each following its own emission spectrum:

$$N_{\mathrm{P}} = N_{0_{\mathrm{P}}} \left(\frac{E}{E_0}\right)^{-2.5} \quad N_{\mathrm{N}} = N_{0_{\mathrm{N}}} \left(\frac{E}{E_0}\right)^{-2.5} \quad N_{\mathrm{I}} = N_{0_{\mathrm{I}}} \left(\frac{E}{E_0}\right)^{-2.2} ,$$

where $E_0 = 10^{17}$ eV, $N_{0_{\mathrm{P}}} = 10^{40}$, $N_{0_{\mathrm{N}}} = 10^{38}$, $N_{0_{\mathrm{I}}} = 10^{39}$ and consider a hard cutoff at $E_C = 2 \cdot 10^{20}$ eV.

i) **(4 points)** Consider only particles with $E > E_0$. What would the spectrum of particles observed at Earth in 1783 be? How many events would be observed now by the Pierre Auger Observatory (area= 3000 km^2)? At which energy? Imagine that the Galactic Magnetic Field between us and the magnetar is uniform and constant, with a direction perpendicular to the galactic plane and its intensity is 1 μG.

Hint: consider $\arcsin x \approx x + \frac{x^3}{6}$.

ii) **(3 points)** Cosmic rays reach the Earth and immediately interact with the atmosphere. In the first interactions Kaons are often produced. Imagine that a K^+ with energy E=1 TeV is produced in the high atmosphere (h=10 km). Which particles would it most likely decay into? Compute the energies in the Earth's reference system of its daughter particles. Are they likely to reach the ground or will they decay again? In which particles? (consider only the most probable decays)

iii) **(3 points)** A muon produced in the high atmosphere (at an altitude h =1000 m) moves towards Earth in a radial direction, on the equatorial plane from the projection of the Greenwich meridian. Suppose that the Earth magnetic field has a value of 0.5 G, that it is constant and that it is oriented towards the North Pole (ignoring the discrepancy between the magnetic North Pole and the geographical one). What momentum is needed for the muon so that it hits the city of Nairobi (approximately longitude 37°E and latitude 0°)? Consider the Earth radius to be 6400 km. If the muon has the momentum you just computed, what is its probability to reach Nairobi, if the average life of a muon is $\tau_\mu = 2 \cdot 10^{-6}$ s?

10 General relativity

Consider a small perturbation of the Minkowski spacetime produced by gravitational waves, $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$.

i) **(0.5 points)** Show that, if n^μ is a null vector in Minkowski, then the vector $s^\mu = n^\mu - \frac{1}{2}\eta^{\mu\rho}h_{\rho\lambda}n^\lambda$ is null in the perturbed metric (always work to $\mathcal{O}(h)$).

ii) **(5 points)** Let us imagine s^μ is the trajectory of a photon emitted at initial frequency ν_0 by a source located at direction $\hat{n}$ from Earth. Compute, using the geodesic equation in the TT gauge, the change in the frequency of the photon, $(\nu(t) - \nu_0)/\nu_0$, produced by a monochromatic gravitational wave propagating along the z direction, $h_{\mu\nu} = h_{\mu\nu}(t - z)$.

iii) **(1 point)** Generalize the result for an arbitrary direction of propagation of the gravitational wave $\hat{\Omega}$.

iv) **(1 point)** What happens for $\hat{\Omega} \cdot \hat{n} = 1$ (*hint: answer this question assuming* $\hat{\Omega} = \hat{z}$)?

v) **(0.5 points)** Is the result linear, *i.e.* additive, for N gravitational waves propagating all in different directions?

vi) **(1 point)** Now suppose that gravitational waves have a modified dispersion relation $\omega(k) = c_s k$ ($c_s = 1$ in General Relativity). How would this new effect manifest in the gravitational redshift derived above?

vii) **(1 point)** In a real observation, gravitational waves are not perfectly chromatic. For example, in a binary system of two object with the same mass m, the frequency f of the gravitational waves evolves with time as

$$\frac{df}{dt} \sim 300 \left(\frac{Gm}{c^3}\right) f^{\frac{11}{3}} \tag{90}$$

during the in-spiraling phase. For $m = 10^{11}\mathrm{M}_\odot$ and $f \sim 1\mathrm{nHz}$, estimate how close to Earth must the source be in order to ignore the frequency evolution of the gravitational wave.